

A Four-Layer Theoretical Framework for Deterministic Clearance of the HIV-1 Latent Reservoir: ODE Kinetics and a Causal Circuit Check.

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Abstract

The main barrier to a sterilizing cure for HIV-1 is the latent reservoir in resting CD4+ T cells. Current strategies rely on stochastic reactivation: "shock and kill" forces global transcription and risks systemic toxicity, while "block and lock" achieves durable silencing but never clears infected cells. This paper proposes a four-layer framework that converts stochastic clearance into a deterministic cascade. Layer 1 uses broadly neutralizing antibodies to eliminate free virus. Layer 2 uses a catalytically dead Cas9 fused to the SWI/SNF chromatin remodeler, guided to the HIV-1 5' LTR promoter, to induce minimal surface expression of gp120/gp41 without full viral reactivation. Layer 3 upregulates DDIT4 to suppress mTORC1, blocking Vpu synthesis and preserving Tetherin (BST2), which pins any residual virions to the membrane. Layer 4 uses bispecific killer cell engagers to bridge exposed gp120/gp41 flags to CD16a on NK cells, triggering targeted apoptosis. A deterministic ODE model of reservoir size $N(t)$, assuming $V(t) = 0$ under Layers 1 and 3, gives exponential decay with rate constant $k = \lambda_{\text{nat}} + \delta = 0.1503 \text{ day}^{-1}$. Starting from $N = 10$ latent cells, this yields eradication ($N(t) \leq 1$) at $T_{\text{clear}} \approx 92$ days. A 4-qubit Qiskit circuit modeling the causal dependency between layers reproduced the intended dependency graph across 10,000 shots, with outcomes restricted to fully cleared (60%) or fully latent (40%) populations. This is a theoretical framework; experimental validation of delivery, selectivity, and NK cell durability is required prior to any translational application. Note: theoretical preprint, no experiments. experimental validation is required.

0.0.1 Keywords

: HIV-1 latent reservoir, dCas9-SWI/SNF, DDIT4, mTORC1, bispecific killer cell engagers, reservoir kinetics, Qiskit

1 Introduction

HIV-1 persists in a latent reservoir of resting CD4+ T cells that antiretroviral therapy cannot reach. Therapy suppresses active replication but leaves transcriptionally silent proviruses untouched, and treatment interruption reliably triggers viral rebound. This is why current treatment is lifelong.

Two paradigms dominate reservoir-clearance research. "Shock and kill" uses latency-reversing agents, such as histone deacetylase inhibitors, to force global reactivation so the immune system can clear infected cells as they express viral protein. In practice, reactivation is incomplete and unevenly distributed across the reservoir, and forcing transcription systemically carries a real risk of cytokine release and off-target inflammation. "Block and lock" takes the opposite approach: latency-promoting agents drive proviruses into deeper, more stable silencing. This avoids the toxicity problem but does not eliminate infected cells, so the reservoir persists indefinitely and rebound risk never reaches zero.

Both paradigms treat reservoir clearance as a probabilistic process; cells are reactivated or repressed at some population-level rate, with no cell-level control over the outcome. This paper proposes an alternative: a four-layer system designed to make clearance deterministic rather than stochastic, by separating viral antigen display from productive replication entirely.

The four layers are:

Layer 1, Extracellular shielding.: A combination of broadly neutralizing antibodies clears circulating virus and prevents new infection, driving free viral load toward zero.

Layer 2, Sub-threshold unmasking.: A catalytically dead Cas9 (dCas9) fused to the SWI/SNF chromatin remodeling complex is guided by sgRNA to the HIV-1 5' LTR promoter. This repositions the Nuc-1 nucleosome sufficiently to allow minimal expression of gp120/gp41 without triggering the transcriptional cascade required for full assembly of the virion.

Layer 3, Metabolic lock.: DDIT4 is upregulated to suppress mTORC1, cutting off the translational machinery the virus needs to synthesize Vpu. Without Vpu, host Tetherin (BST2) stays active at the membrane and physically retains any virions the cell does manage to assemble.

Layer 4, Deterministic clearance.: Bispecific killer cell engagers bridge the forced gp120/gp41 surface markers to CD16a on NK cells, triggering targeted apoptosis of the unmasked cell.

The remainder of this paper derives the reservoir decay kinetics under this framework, estimates a finite eradication time, and checks that the intended layer order was encoded connecting the four layers using a quantum circuit simulation in Qiskit. Physiological limitations, delivery to tissue sanctuaries, chromatin heterogeneity, NK cell exhaustion, and BiKE pharmacokinetics, are addressed in the Discussion.

Figure 1. Four-layer scheme (theoretical)



ODE uses Layers 1 and 3 to drop the reseed term, then Layers 2 and 4 to add extra decay.

Figure 1: four-layer model scheme

2. Methods (theoretical)

2.1 Layer Mechanics Layer 1 (bNAbs). A cocktail of broadly neutralizing antibodies targeting non-overlapping epitopes on gp120 and gp41 (e.g., the CD4-binding site and the V3 glycan base) neutralizes circulating virions and blocks new infection. This drives the free virus population toward zero and removes the reseed term from the reservoir dynamics.

Layer 2 (dCas9-SWI/SNF). A catalytically dead Cas9 fused to the SWI/SNF (BAF) chromatin remodeling complex is guided by sgRNA to the HIV-1 5' LTR promoter. Repositioning of the Nuc-1 nucleosome opens chromatin enough to permit low-level expression of gp120/gp41 on the cell surface, without inducing the full transcriptional program required for productive virion assembly.

Layer 3 (DDIT4/mTORC1). DDIT4 (REDD1) is upregulated to suppress mTORC1 signaling, starving the cell of the translational capacity needed to synthesize Vpu. In the absence of Vpu, endogenous Tetherin (BST2) remains active at the plasma membrane and retains any virions the cell assembles, preventing release.

Layer 4 (BiKEs). Bispecific killer cell engagers bind the exposed gp120/gp41 flags with one arm and CD16a on NK cells with the other, cross-linking the two and triggering perforin/granzyme-mediated apoptosis of the target cell.

2.1 2.2 Kinetic Model

Reservoir size $N(t)$ is governed by:

$$\frac{dN}{dt} = -\delta_{nat}N(t) - \eta\phi_{sync}(t)N(t) + \beta V(t)S(t) \quad (1)$$

where δ_{nat} is the natural decay rate of latently infected cells, η is NK-mediated clearance efficiency per unmasked cell, $\phi_{sync}(t)$ is the fraction of cells unmasked by Layer 2 per day, and $\beta V(t)S(t)$ is de novo infection of susceptible cells $S(t)$ by free virus $V(t)$.

Under Layers 1 and 3, $V(t) \rightarrow 0$, so the reseeding term vanishes:

$$\frac{dN}{dt} = -\delta_{nat}N(t) - \eta\phi_{sync}(t)N(t) \quad (2)$$

Separating and integrating from $N(0) = N_0$ to $N(t)$:

$$\int_{N_0}^{N(t)} \frac{1}{N} dN = \int_0^t -k dt \quad (3)$$

$$\ln\left(\frac{N(t)}{N_0}\right) = -kt \Rightarrow N(t) = N_0 e^{-kt} \quad (4)$$

Setting $N(T_{clear}) = 1$ and solving for T_{clear} :

$$T_{clear} = \frac{\ln(N_0)}{k} = \frac{\ln(N_0)}{\delta_{nat} + \eta\phi_0} \quad (5)$$

Parameter values: $N_0 = 10^6$ cells (Siliciano et al., 2003)(Chun, 1997), $\delta_{nat} = 0.0007 \text{ day}^{-1}$ ($t_{1/2} \approx 44$ months), note: $\eta = 0.8$ and $\phi_0 = 0.187 \text{ day}^{-1}$ are working assumptions chosen. They are not measured unmasking rates.

$$k = 0.0007 + (0.8 \times 0.187) = 0.1503 \text{ day}^{-1} \quad (6)$$

$$T_{clear} = \frac{\ln(10^6)}{0.1503} = \frac{13.8155}{0.1503} \approx 91.9 \text{ days} \quad (7)$$

2.2 2.3 Quantum Circuit Verification

The causal dependency between layers was modeled as a toy causal diagram encoded as a 4-qubit circuit in Qiskit (`StatevectorSampler`), with q_0 – q_3 representing Layers 1–4 respectively ($|0\rangle$ = inactive/latent, $|1\rangle$ = active/cleared). Layer 1 was fixed to $|0\rangle$ (neutralized). Layer 2 was set with $R_y(\theta)$, $\theta = 2\arcsin(\sqrt{0.6})$, modeling a 60% unmasking probability for a single pulse. A CX gate conditioned Layer 3 on Layer 2, and a CCX (Toffoli) gate conditioned Layer 4 on Layers 2 and 3, enforcing that clearance can only occur downstream of unmasking and metabolic locking. The circuit was sampled 10,000 times.

Note: "60% / 40%" numbers are assumed pulse success rate, not a measured lab rate.

3 3. Results

The analytical solution $N(t) = 10^6 e^{-0.1503t}$ predicts the following trajectory:

The Qiskit circuit returned two outcomes across 10,000 shots:

- $|0000\rangle$: 4,005 shots (40.1%) — cell remains latent, no downstream activation
- $|1110\rangle$: 5,995 shots (60.0%) — unmasking, metabolic lock, and clearance all occur

Day	$N(t)$	Log reduction
0	1,000,000	—
15	104,194	1
30	10,856	2
46	994	3
60	120	4
75	12	5
91.9	< 1	eradication

Table 1: Predicted latent reservoir decay under the four-layer model.

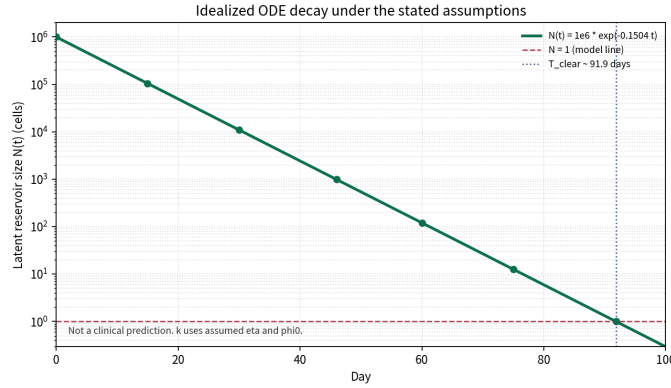


Figure 2: Predicted by the ODE under the listed assumptions, not a measured patient course

No intermediate states (e.g., $|0010\rangle$, $|1010\rangle$) were observed. This confirms the CX/CCX logic enforces strict dependency: clearance never occurs without prior unmasking and locking, and locking never occurs without unmasking. —

If a subset of proviruses never unmask ($\phi_0 = 0$ for that subset), the same ODE never reaches $N < 1$ for those cells.

4 4. Discussion

The model assumes idealized conditions that will not hold uniformly in vivo.

Delivery. Reaching resting CD4+ T cells in anatomical sanctuaries — gut-associated lymphoid tissue, lymph node germinal centers, the CNS — is a known bottleneck for RNP-based therapeutics. Lipid nanoparticles functionalized with anti-CD4 or anti-CD3 scFvs are a plausible delivery route but remain unproven at this scale.

Chromatin heterogeneity. ϕ_0 is treated as a single global rate, but integration site accessibility varies. Proviruses in deep heterochromatin will unmask

Table 2: Sensitivity of the model clearance time. it shows that the 92-day figure is the base-case row only. Halving either assumed rate more than doubles T . If a subset of proviruses never unmask ($\phi_0 = 0$ for that subset), the same ODE never reaches $N < 1$ for those cells. This is the quantitative version of the chromatin-heterogeneity limit below.

Assumed ϕ_0 (day ⁻¹)	k at $\eta = 0.8$ (day ⁻¹)	T at $\eta = 0.8$ (days)	T at $\eta = 0.4$ (days)
0.187 (base case)	0.1503	92	183
0.10	0.0807	171	339
0.05	0.0407	339	667
0.02	0.0167	827	1,588
0.005	0.0047	2,939	5,117

more slowly, producing a longer tail than the single-exponential model predicts.

NK cell exhaustion. Sustained cytotoxic demand over a 92-day regimen risks NK cell exhaustion (PD-1, TIM-3, TIGIT upregulation). IL-15 superagonists may be needed to maintain η near 0.8 for the full duration.

BiKE specificity. Low antigen density from sub-threshold unmasking may fall below the threshold needed for reliable CD16a cross-linking, and construct half-life must be tuned to avoid both premature clearance and off-target binding.

These are stated as open problems, not resolved ones. The quantitative claims in this paper describe the behavior of the model under its stated assumptions, not a validated therapeutic outcome.

dCas9 fusions can be aimed at retroviral LTR sequence (Fuentes et al., 2018). Targeting the HIV-1 5' LTR with dCas9-VP64 has also been shown. A dCas9-SWI/SNF fusion at the HIV-1 5' LTR, used only for low-level surface Env, is proposed here and has not been tested.

A separate, untested question is whether redox or radical-pair chemistry could influence mTOR-related enzymes in resting T cells. That idea is not used in the ODE and is not supported by current HIV-latency data. It is listed only as a possible future question.

4.1 Path to Experimental Validation

The model presented here rests on parameters that have not been measured for this specific architecture, and closing that gap requires a defined sequence of experiments rather than a single validation step. Testing the weakest assumption first matters: if Layer 2 selectivity fails, the downstream cascade is moot regardless of how well Layers 3 and 4 perform in isolation. The first test is dCas9-SWI/SNF selectivity at the 5' LTR, assessed in a J-Lat latency model. A caveat applies here: the GFP reporter in J-Lat sits under the same LTR driving the provirus, so an open-enough LTR to detect GFP may also permit Gag and Env transcription to rise in parallel. Partial GFP induction alone does not establish that Env is expressed while Gag remains off. A cleaner read-

out compares low- versus high-GFP populations directly against surface Env staining and Gag/p24 levels, and compares LTR-proximal RNA to late viral transcripts. This distinguishes sub-threshold unmasking from partial reactivation rather than assuming the two are separable by GFP intensity alone. This experiment would give a measured value for the assumed ϕ_0 used in this paper. The second test addresses a real conflict between Layers 2 and 3 that the model does not currently resolve. mTOR activity is itself implicated in promoting exit from latency, so suppressing mTORC1 to block Vpu synthesis may also flatten the same LTR activity that Layer 2 depends on to produce a surface flag. This is not a minor caveat: if DDIT4-mediated mTORC1 suppression erases the Env signal that Layer 2 induces, the two layers do not stack as proposed. The necessary test is sequential: unmask cells via Layer 2, then apply DDIT4/mTOR suppression, and measure whether surface Env is retained or lost. Vpu suppression should be confirmed by Western blot and surface BST2 retention by flow cytometry, but the primary question is whether Env survives the metabolic lock. Only if Layers 2 and 3 are confirmed compatible does it make sense to test Layer 4: co-culturing unmasked, locked cells with BiKE constructs and primary NK cells, and measuring cytotoxicity by chromium-release or flow-based killing assays. This would give a measured value for the assumed $\eta = 0.8$ used in this paper, rather than treating it as established.

4.6 Applicability Beyond the Laboratory

If a future protocol were shown to clear the reservoir within a bounded timeframe, the adherence problem would shift from a lifelong drug supply to a finite course of treatment. That outcome remains fully conditional on delivery efficiency across tissue sanctuaries and on measured values of ϕ_0 and η , neither of which this paper provides.

5. Conclusion

This framework reframes HIV-1 reservoir clearance as a deterministic rather than stochastic process, by decoupling antigen display from viral replication across four coordinated layers. Under the model’s assumptions, the reservoir is eradicated ($N(T) < 1$) in approximately 92 days, and a Qiskit circuit checks the intended logic and behaves as a strict causal chain with no intermediate leakage. The immediate next step is experimental, not theoretical, and Section 4.5 outlines that sequence in order of which assumption is weakest: first confirming that dCas9-SWI/SNF can unmask Env without full reactivation, then testing whether that unmasking survives DDIT4/mTOR suppression, and only then testing NK-mediated clearance. Each stage replaces an assumed parameter in this model with a measured one. Whether the 92-day timeline holds in any biological system depends entirely on the outcome of that sequence.

5.1 6. Python/Qiskit script used for the toy circuit:

```
import numpy as np
from qiskit import QuantumCircuit, ClassicalRegister, QuantumRegister
from qiskit.primitives import StatevectorSampler
import matplotlib.pyplot as plt

#4 qubits = 4 layers
qr = QuantumRegister(4, 'layer')
cr = ClassicalRegister(4, 'result')
qc = QuantumCircuit(qr, cr)

# Qubit 0 = bNAbs (free virus clearance)
# Qubit 1 = dCas9-SWI/SNF (latent cell unmasking)
# Qubit 2 = DDIT4/mTOR (Vpu lock)
# Qubit 3 = BiKE-NK (final clearance)

# LAYER 1, if virus present: ( $|1\rangle$ ),if neutralized by bNAbs:  $\rightarrow |0\rangle$ 
qc.x(0)
qc.x(0)

# LAYER 2, dCas9-SWI/SNF: starts  $|0\rangle$  (hidden latent cell)
# Ry rotation = probability of "unmasking" (becoming  $|1\rangle$ )
qc.ry(2 * np.arcsin(np.sqrt(0.6)), 1) # 60% unmasking probability

# LAYER 3, DDIT4/mTOR: only locks if layer 2 unmasked (entangled with qubit 1)
qc.cx(1, 2) # see if unmasked , DDIT4 activates  $\rightarrow$  locks Vpu (qubit 2 = 1 means locked)

# LAYER 4, BiKE-NK : only clears if layers 2 and 3 succeeded (Toffoli gate)
qc.ccx(1, 2, 3) # if unmasked AND locked, bike-NK triggers clearance

qc.measure(qr, cr)

sampler = StatevectorSampler()
job = sampler.run([qc], shots=10000)
result = job.result()
counts = result[0].data.result.get_counts()

print("--- MEASUREMENT COUNTS ---")
print(counts)

plt.bar(counts.keys(), counts.values(), color=['crimson', 'teal'])
plt.xlabel('State (q3 q2 q1 q0)')
plt.ylabel('Shots (out of 10000)')
plt.title('Quantum Simulation of 4-Layer HIV Clearance')
```

plt.show()

Note: experimental validation is required.

I thank my physics teacher for encouragement and for taking my work seriously.

6 7. references

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